**Project Design Phase**
**Problem – Solution Fit**

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|---|---|
| Date | 1 July 2026 |
| Team ID | |
| Project Name | Rising Waters |
| Maximum Marks | 2 Marks |

**Problem – Solution Fit**

The **Rising Waters** project addresses the challenge of predicting floods accurately using historical weather and rainfall data. Floods often occur unexpectedly, causing loss of life, property damage, and disruption to communities. Traditional methods may not provide timely and reliable flood warnings.

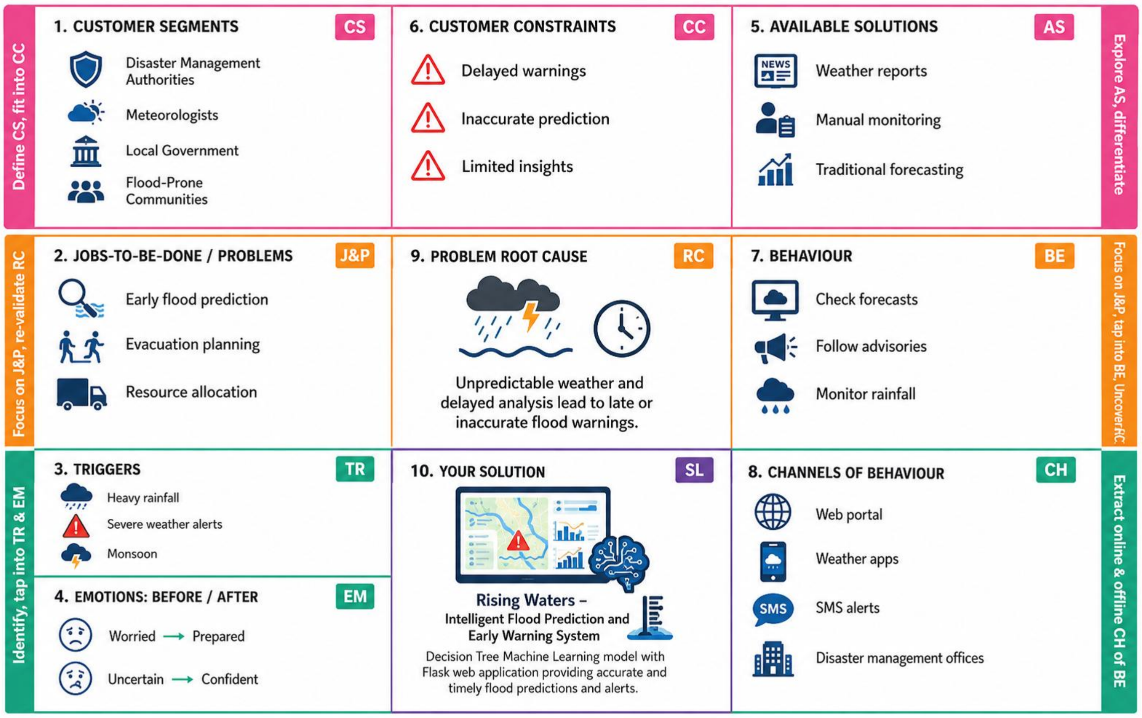
Our solution uses a **Machine Learning-based Decision Tree model** to analyze weather parameters and predict the likelihood of flooding. Through a simple Flask web application, users can enter weather data and instantly receive flood risk predictions along with safety recommendations.

The system is designed to help disaster management authorities, meteorologists, and communities make informed decisions, improve disaster preparedness, and support early evacuation planning.

**Purpose:**

- ❑ Predict flood risk accurately using Machine Learning.
- ❑ Provide early warnings to support disaster preparedness.
- ❑ Help disaster management authorities make faster and better decisions.
- ❑ Improve evacuation planning and resource allocation during emergencies.
- ❑ Provide a simple, user-friendly web application for flood risk prediction.
- ❑ Support future enhancements such as real-time weather API integration and cloud deployment.
- ❑ Improve public safety by reducing the impact of flood disasters
- ❑ **Improve public safety by reducing the impact of flood disasters.**

**Template:**



Define CS, fit into CC
1. CUSTOMER SEGMENTS
CS
Disaster Management Authorities
Meteorologists
Local Government
Flood-Prone Communities
6. CUSTOMER CONSTRAINTS
CC
Delayed warnings
Inaccurate prediction
Limited insights
5. AVAILABLE SOLUTIONS
AS
NEWS
Weather reports
Manual monitoring
Traditional forecasting
Explore AS, differentiate
Focus on J&P, re-validate RC
2. JOBS-TO-BE-DONE / PROBLEMS
J&P
Early flood prediction
Evacuation planning
Resource allocation
9. PROBLEM ROOT CAUSE
RC
Unpredictable weather and delayed analysis lead to late or inaccurate flood warnings.
7. BEHAVIOUR
BE
Check forecasts
Follow advisories
Monitor rainfall
Focus on J&P, tap into BE, Uncover RC
Identify, tap into TR & EM
3. TRIGGERS
TR
Heavy rainfall
Severe weather alerts
Monsoon
4. EMOTIONS: BEFORE / AFTER
EM
Worried → Prepared
Uncertain → Confident
10. YOUR SOLUTION
SL
Rising Waters –
Intelligent Flood Prediction and Early Warning System
Decision Tree Machine Learning model with Flask web application providing accurate and timely flood predictions and alerts.
8. CHANNELS OF BEHAVIOUR
CH
Web portal
Weather apps
SMS
SMS alerts
Disaster management offices
Extract online & offline CH of BE